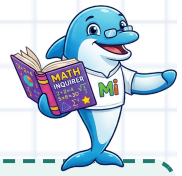
# COPY ON THE GRID!

Name _____ Date _____

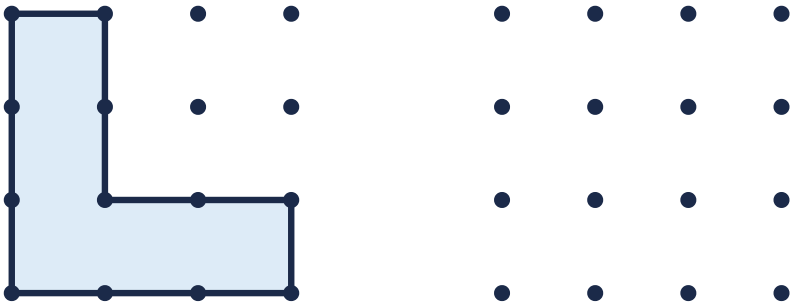


After question 3 you may choose: carry on, or try **Think harder**.

*Count the steps between the dots, not the dots. Say it out loud, then draw or write it.*

## No.S1

**Mei draws an L shape on a grid.**

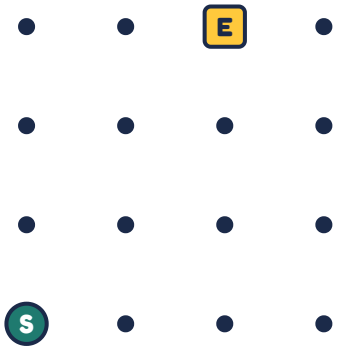


**(a)** Copy the L shape on the empty grid.

**(b)** The L shape is ☐ steps across and ☐ steps up.

## No.S2

**A counter starts at S. It moves only right and up to reach E.**

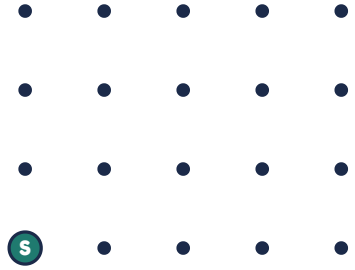



**(a)** The counter moves ☐ right and ☐ up.

**(b)** Draw the counter's path from S to E.

**No.S3**

**Start at S. Go 4 steps right. Then go 1 step up.**




**(a)** Draw the path.

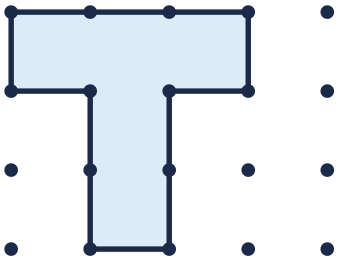
**(b)** Draw a ring round the dot where you stop.

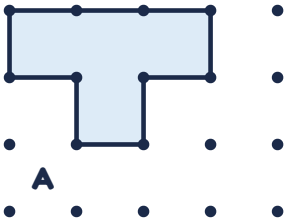
Name _____ Date _____

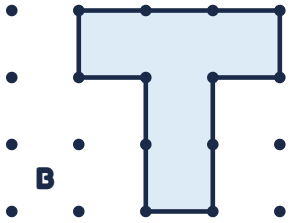
*Use your grid page and a button. A ruler or the edge of a card helps with straight lines.*

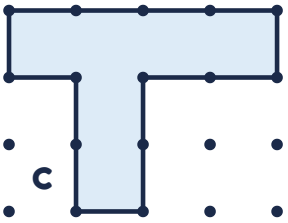
## No.S4

**Which picture is an exact copy of the T shape?**





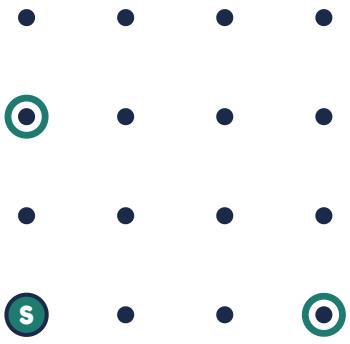




Circle A, B or C. A B C

## No.S5

**Three corners of a rectangle are ringed. Join the dots to finish the rectangle.**

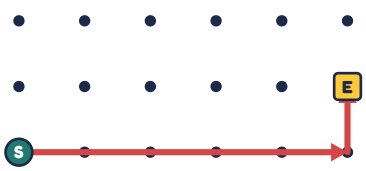



**(a)** Draw a ring round the missing corner.

**(b)** The missing corner is ☐ right and ☐ up from S.

## No.S6

**Look at the path from S to E. Tick True or False for each sentence.**




| Statement | True | False |
|---|---|---|
| The path goes 5 steps right. | ☐ | ☐ |
| The path goes 2 steps up. | ☐ | ☐ |
| The bottom line touches 5 dots. | ☐ | ☐ |
| E is 5 right and 1 up from S. | ☐ | ☐ |

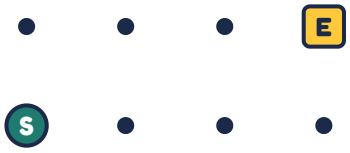
Put a tick (✓) in the correct box.

Name _____ Date _____

*Think about every route. List them in order so none is missed. Choose any question, in any order.*

## No.T1

**A counter goes from S to E. It moves only right or up. The grid is 3 steps across and 1 step up.**

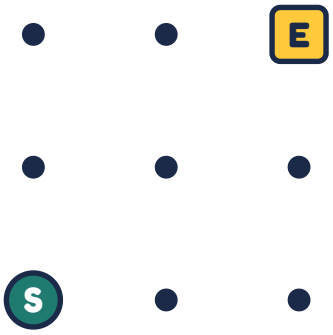



**(a)** One route is R R R U. R means right and U means up. Write the other routes: _____

**(b)** There are ☐ routes in all.

## No.T2

**A counter goes from S to E. It moves only right or up. The grid is 2 steps across and 2 steps up.**

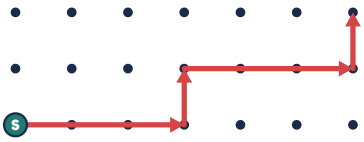



**(a)** Three routes start with U: U U R R, U R U R and U R R U. Write the routes that start with R: _____

**(b)** There are ☐ routes in all.

## No.T3

**Ravi's counter walks 3 right and 1 up, again and again. The picture shows 2 sets.**




**(a)** After 3 sets, the counter is ☐ right and ☐ up from S.

**(b)** After 5 sets, the counter is ☐☐ right and ☐ up from S.

## No.T4

**Start at S. Go 3 steps right. Go 2 steps up. Then go 1 step left.**




**(a)** Draw the path.

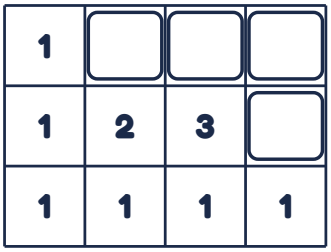
**(b)** You stop ☐ right and ☐ up from S.

Name _____ Date _____

*Look at the boxes that are filled in and see how they are made. Choose any question. You do not have to finish them all.*

## No.G1

**The boxes show how many routes reach each corner. S is the bottom left box. E is the top right box.**

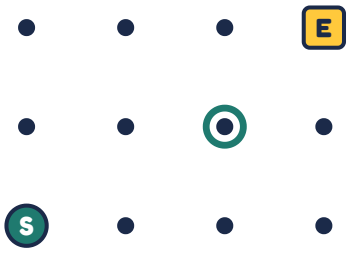



**(a)** Fill in the empty boxes.

**(b)** There are ☐☐ routes from S to E.

## No.G2

**A counter goes from S to E, moving only right or up. The grid is 3 steps across and 2 steps up. The route must pass the ringed corner.**



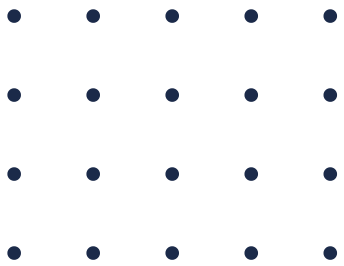

**(a)** Routes from S to the ringed corner: ☐

**(b)** Routes from the ringed corner to E: ☐

**(c)** Routes from S to E that pass the ringed corner: ☐

**No.G3**

**Draw a shape with 4 corners on the grid. A is its bottom left corner. Many answers are right.**



**(a)** B is ☐ right and ☐ up from A.

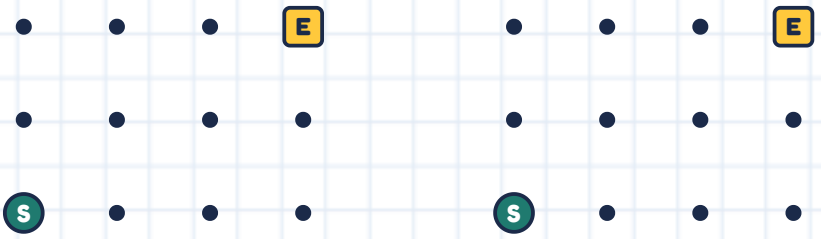
**(b)** C is ☐ right and ☐ up from A.

**(c)** D is ☐ right and ☐ up from A.

Name _____ Date _____

## Investigation: Same steps, different routes

**First:** Draw two different routes from S to E, one on each grid. Count the steps in each route: _____ steps and _____ steps.



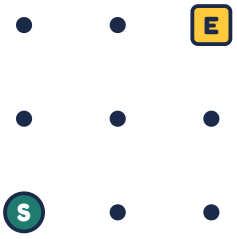

**Next:** Do all the routes from S to E have the same number of steps? Write why.

## Exit ticket

*On your own, at the end of the lesson. No looking back.*

### No.E1

**A counter starts at S and moves only right and up to reach E.**




(a) The counter moves ☐ right and ☐ up.

### No.E2

**A counter moves 2 right and 3 up. Then it moves 3 right and 1 up.**

(a) In all it moves ☐ right and ☐ up.

**No.E3**

**A figure is 5 steps across and 2 steps up. A child draws a copy that is 6 steps across and 2 steps up.**

**(a)** Is it an exact copy? Circle Yes or No. **Yes** **No**

**(b)** An exact copy is ☐ steps across.

**Look-back** There are no right answers. You can pass. Nobody reads it out.

**Something I did well:**

**The hardest part was ...** ☐ counting steps, not dots ☐ copying exactly ☐ counting the routes ☐ something else

**What helped me:** ☐ the grid page ☐ my button ☐ drawing the routes ☐ I was not stuck

**Next time I will try:** ☐ counting steps ☐ saying where to go ☐ finding every route ☐ asking a question

☐ I am still stuck (my grown-up will photograph this page for my tutor)

☐ I would like to pass

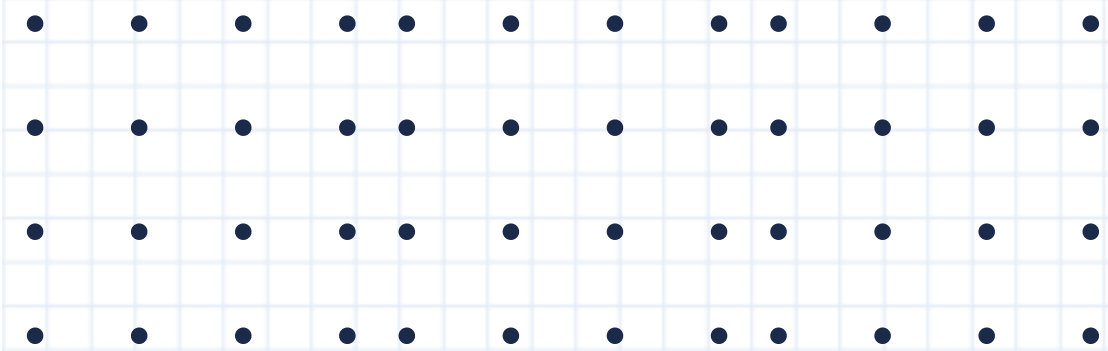
Name _____ Date _____

# Kit page: grid page

No cutting: this page is for drawing on. The three small grids are route grids for walking a button. The two wide grids are copy grids: the dashed line splits each one, so copy the figure from the screen into the right-hand half. No printer? Draw rows of dots on paper.

**No cutting needed: keep this page flat to use in the lesson. Print it on its own sheet, or on one side only.**

## Grids: keep this page flat to draw on. No cutting.



## Copy grids: draw the other half

